Lecture 5

Readings on intervals: spectral measures

Objectives. *By the spectral theorem, a cited result, self-adjoint operators correspond to spectral measures. We find the spectral measure explicitly for the position and the momentum and for every self-adjoint operator with an eigenbasis. The momentum with walls lies inside the integral of a spectral measure for every boundary condition, so a symmetric operator need not decide its readings.*

- Compute an interval projection and its probability.
- Apply the spectral-measure update after a yes.
- Explain why no state on the line has both its position and its momentum in bounded intervals.

Lecture 5, first part

Spectral measures and the spectral theorem

5.1 Spectral measures

Three rules for readings have appeared so far: the projections E_a of an eigenbasis (Postulate 1.5), the projections 1_I of the position in a box (Postulate 2.6), and the projections $F^{-1}1_I F$ of the momentum on a line (Postulate 4.5). In each rule a projection belongs to a set of readings, and sets that exclude each other have orthogonal projections. This section collects what the three rules share.

In this lecture an *interval* is a set of real numbers of one of the forms (a, b), $[a, b]$, $(a, b]$ or $[a, b)$, with $a \le b$. An endpoint may be $-\infty$ or $+\infty$ when its bracket is round. So $\mathbb{R}$ is an interval, and so are a single point $[a, a]$ and the empty set (a, a).

Definition 5.1 (Spectral measure)**.** A *spectral measure* on a Hilbert space $\mathcal{H}$ assigns to each interval I an orthogonal projection $E(I)$ on $\mathcal{H}$ such that

1. $E(\emptyset) = 0$ and $E(\mathbb{R}) = \mathbb{1}$;
2. $E(I \cap J) = E(I)E(J)$ for all intervals I and J;
3. $E(I \cup J) = E(I) + E(J)$ when I and J are disjoint and $I \cup J$ is an interval;
4. if $I_1 \subseteq I_2 \subseteq \cdots$ are intervals with the union I, then $E(I_n)\psi \to E(I)\psi$ for every $\psi \in \mathcal{H}$.

For disjoint intervals I and J, normalisation and the intersection identity give $E(I)E(J) = E(\emptyset) = 0$. So the ranges of $E(I)$ and $E(J)$ are orthogonal, and readings that exclude each

other have orthogonal projections, as in *Quantum Mechanics I*. For a vector $\psi \in \mathcal{H}$, put

$$\mu_\psi(I) = \langle \psi, E(I)\psi \rangle = \|E(I)\psi\|^2 . \tag{5.1}$$

For the box Hamiltonian, an interval selects the levels it contains. The interval $[3E_1, 10E_1]$ contains $E_2 = 4E_1$ and $E_3 = 9E_1$, so its projection is $|u_2\rangle\langle u_2| + |u_3\rangle\langle u_3|$. On $\psi = (u_1 + u_2)/\sqrt{2}$ its coefficients $c_n = \langle u_n, \psi \rangle$ give

$$\mu_\psi([3E_1, 10E_1]) = |c_2|^2 + |c_3|^2 = \tfrac{1}{2}.$$

For a unit vector ψ, normalisation gives $\mu_\psi(\mathbb{R}) = 1$. By additivity and the orthogonality, μ_ψ adds over two disjoint intervals, and by continuity on increasing intervals, $\mu_\psi(I_n) \to \mu_\psi(I)$. So μ_ψ is a probability distribution on the intervals, the *distribution* of ψ under E.

One function records a distribution. For a unit vector ψ, the *distribution function* is $F_\psi(\lambda) = \mu_\psi((-\infty, \lambda])$; the subscript keeps it apart from the Fourier transform F. It is nondecreasing. It tends to 1 as $\lambda \to \infty$ and, since the intervals $(-n, \infty)$ increase to $\mathbb{R}$, to 0 as $\lambda \to -\infty$. It is continuous from the right. Since the intervals $(\lambda + 1/n, \infty)$ increase to (λ, ∞), normalisation, additivity and continuity on increasing intervals give, on every vector,

$$E((-\infty, \lambda + 1/n]) = \mathbb{1} - E((\lambda + 1/n, \infty)) \to \mathbb{1} - E((\lambda, \infty)) = E((-\infty, \lambda]).$$

So the projections $E((-\infty, \lambda])$ form what Amrein, §4.2.1, calls a *spectral family*. For $a < b$, the probability of $(a, b]$ is $F_\psi(b) - F_\psi(a)$. A point λ with $\mu_\psi([\lambda, \lambda]) > 0$ is a jump of F_ψ, and where the distribution has a continuous density, the density is the slope of F_ψ. Figure 5.1 shows both, and a distribution with both parts.

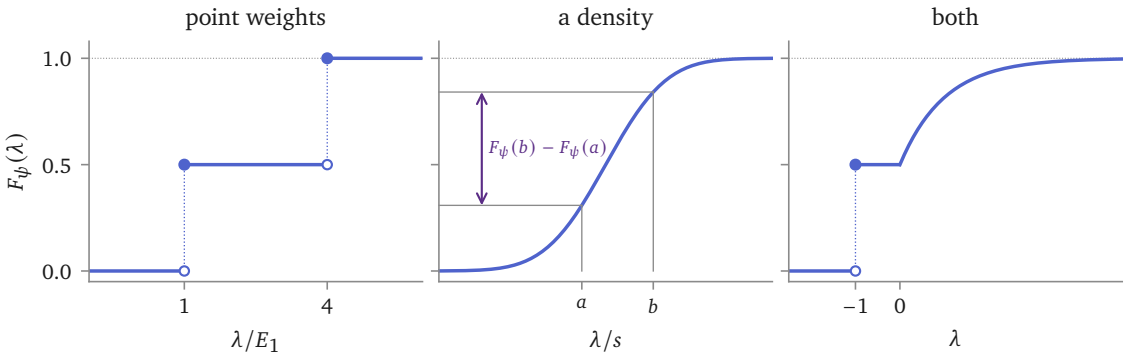


Figure 5.1: Three distribution functions F_ψ. Left: the box energy on $(u_1 + u_2)/\sqrt{2}$, with jumps $\frac{1}{2}$ at E_1 and $4E_1$. Middle: the position on a Gaussian of spread s. The probability of $(a, b]$ is the rise of F_ψ. Right: the distribution of Example 5.2. At a jump the filled dot marks the value of F_ψ.

A measure adds over countably many disjoint sets. The clauses ask for less, but they are enough to give such a measure, by a cited theorem. By Amrein's (4.25)–(4.26), cited, the spectral measure extends to the *Borel sets*, the smallest family of sets of real numbers that contains every interval, the complement of each of its sets and the union of every sequence of its sets. By his (4.29) and continuity on increasing intervals the extension takes the value $E(I)$ on every interval. The extension adds over countably many disjoint sets S_j, in the sense that $E(\bigcup_j S_j)\psi = \sum_j E(S_j)\psi$ for every ψ. We put $\mu_\psi(S) = \|E(S)\psi\|^2$ for a Borel set S.

The integral against μ_ψ weights a function by the distribution. Let f be continuous on a closed bounded interval $[a, b]$. Its integral over $(a, b]$ is the limit of the sums $\sum_j f(\lambda_j)\big(F_\psi(\lambda_j) -$

$F_\psi(\lambda_{j-1})\big)$ as the largest cell of the partition $a = \lambda_0 < \cdots < \lambda_m = b$ shrinks to 0. The limit exists, since f changes little within a small cell and the weights add up to at most 1.

The same idea works with sets in place of cells, and it covers every function we need. Let $f \geq 0$ be piecewise continuous, or the indicator of a Borel set. Then $\int f \,\mathrm{d}\mu_\psi$ is the supremum of the sums $\sum_j c_j \mu_\psi(S_j)$ over finitely many disjoint Borel sets S_j and numbers $c_j \geq 0$ with $\sum_j c_j 1_{S_j} \leq f$. We call each such sum a *lower sum* of f. The supremum may be $+\infty$. In particular, $\int 1_S \,\mathrm{d}\mu_\psi = \mu_\psi(S)$. For a real f with $\int |f| \,\mathrm{d}\mu_\psi < \infty$, the integral of f is the integral of its positive part minus that of its negative part. A complex function is integrated through its real and imaginary parts.

The definition gives the two cases that this lecture computes. If $\mu_\psi(I) = \sum_{k:\,\lambda_k \in I} w_k$ for every interval I, for points λ_k with weights $w_k \geq 0$, then

$$\int f \,\mathrm{d}\mu_\psi = \sum_k f(\lambda_k) w_k.$$

If $\mu_\psi(I) = \int_I w(\lambda)\,\mathrm{d}\lambda$ for every interval I, for a density w, then

$$\int f \,\mathrm{d}\mu_\psi = \int f(\lambda) w(\lambda)\,\mathrm{d}\lambda.$$

If the distribution has both parts, the two contributions add.

Example 5.2 (A distribution with both parts)**.** Under a spectral measure E, let a unit vector ψ have the weight $\frac{1}{2}$ at $\lambda = -1$ and the density $\frac{1}{2}e^{-\lambda}$ on $(0, \infty)$.

1. *The distribution function.* It is 0 for $\lambda < -1$, $\frac{1}{2}$ for $-1 \leq \lambda \leq 0$ and $1 - \frac{1}{2}e^{-\lambda}$ for $\lambda > 0$.
2. *The mean square.* The point weight and the density give

$$\int \lambda^2 \,\mathrm{d}\mu_\psi = \tfrac{1}{2}(-1)^2 + \tfrac{1}{2}\int_0^\infty \lambda^2 e^{-\lambda}\,\mathrm{d}\lambda = \tfrac{1}{2} + 1 = \tfrac{3}{2},$$

   which is finite.
3. *One lower sum.* Take $f(\lambda) = \lambda^2$. Give the set $\{-1\}$ the height $f(-1) = 1$. Cut $(0, R]$ into cells $(a_j, b_j]$ and give each the height a_j^2, which lies below f on the cell, since $a_j \geq 0$. A cell has the probability $\mu_\psi((a_j, b_j]) = \frac{1}{2}(e^{-a_j} - e^{-b_j})$, so the lower sum is

$$\tfrac{1}{2} + \sum_j \frac{a_j^2}{2}\big(e^{-a_j} - e^{-b_j}\big).$$

   For $R = 2$ and four cells of length $\frac{1}{2}$ it is 0.701, well below $\frac{3}{2}$.
4. *Refining.* Let every cell have length at most h. Since $\frac{1}{2}a_j^2(e^{-a_j} - e^{-b_j}) = \frac{1}{2}\int_{a_j}^{b_j} a_j^2 e^{-\lambda}\,\mathrm{d}\lambda$ and $\lambda^2 - a_j^2 = (\lambda + a_j)(\lambda - a_j) \leq 2Rh$ on the cell,

$$0 \leq \frac{1}{2}\int_0^R \lambda^2 e^{-\lambda}\,\mathrm{d}\lambda - \sum_j \frac{a_j^2}{2}\big(e^{-a_j} - e^{-b_j}\big) \leq Rh \int_0^R e^{-\lambda}\,\mathrm{d}\lambda \leq Rh.$$

   As $R \to \infty$, the first term tends to $\frac{1}{2}\int_0^\infty \lambda^2 e^{-\lambda}\,\mathrm{d}\lambda = 1$. So a large R and then a small h bring the lower sums as close to $\frac{3}{2}$ as we like. For a fixed R they stay below $\frac{1}{2} + \frac{1}{2}\int_0^R \lambda^2 e^{-\lambda}\,\mathrm{d}\lambda < \frac{3}{2}$, so the cells must also reach further out.

A state of a square well with both a bound and a continuum component has an energy distribution with a point part and a continuous part (Lecture 12).

Proposition 5.3 (Spectral measures from an eigenbasis and from indicators)**.**

1. From an eigenbasis. *Let (e_k) be an orthonormal basis of $\mathcal{H}$, with k running through $\mathbb{N}$ or through $\mathbb{Z}$, and let the λ_k be real. Then*
$$E(I)\psi = \sum_{k:\,\lambda_k \in I} \langle e_k, \psi\rangle\; e_k$$
defines a spectral measure.
2. Indicators. *On $L^2(\mathbb{R})$, the projections 1_I of Definition 2.5, now on the line, form a spectral measure, and so do the projections $U^{-1}1_I U$, for every unitary U.*

Proof. In both parts, normalisation, the intersection identity and additivity follow directly, and only continuity on increasing intervals needs an estimate.

1. Write $c_k = \langle e_k, \psi\rangle$. Since $\sum_k |c_k|^2 = \|\psi\|^2$ by Parseval's identity, the series converges, and $E(I)$ is the orthogonal projection on the closed span of the e_k with $\lambda_k \in I$. Normalisation holds, since $E(\emptyset)\psi$ is the empty sum and $E(\mathbb{R})\psi = \sum_k c_k e_k = \psi$ is the expansion of ψ in the basis. The intersection identity and additivity hold because $E(I)$ keeps exactly the terms with $\lambda_k \in I$. For continuity on increasing intervals,
$$\|E(I)\psi - E(I_n)\psi\|^2 = \sum_{k:\,\lambda_k \in I\setminus I_n} |c_k|^2 \,.$$
Given $\epsilon > 0$, choose a finite set K of indices with $\sum_{k\notin K} |c_k|^2 < \epsilon$. Each $\lambda_k \in I$ lies in some I_n, and then in all later ones. So for large n no $k \in K$ has $\lambda_k \in I \setminus I_n$, and the sum is less than ϵ.
2. As in § 2.4, each 1_I is an orthogonal projection. Normalisation, the intersection identity and additivity are the identities $1_\emptyset = 0$, $1_{\mathbb{R}} = 1$, $1_{I\cap J} = 1_I 1_J$ and $1_{I\cup J} = 1_I + 1_J$ for disjoint I and J. For continuity on increasing intervals, let $\psi \in L^2(\mathbb{R})$ and $\epsilon > 0$. By the standing fact S4, there is a continuous f that vanishes outside some $[-R, R]$, with $\|\psi - f\| < \epsilon$. Let M bound $|f|$. If I is empty there is nothing to prove. Otherwise let $a \le b$ be the endpoints of I and, leaving out the empty I_n, let $a_n \le b_n$ be those of I_n. Then $a_n \to a$ and $b_n \to b$, and $I \setminus I_n$ lies in the closed intervals from a to a_n and from b_n to b, which are half-lines when a or b is infinite. Let ℓ_n be the length of the part of these intervals inside $[-R, R]$. Since $a_n \to a$ and $b_n \to b$, the length ℓ_n tends to 0. Since $1_I - 1_{I_n} = 1_{I\setminus I_n}$, the triangle inequality gives the first step below. The projection $1_{I\setminus I_n}$ does not increase norms, and $|f| \le M$ vanishes outside $[-R, R]$, which gives the second:
$$\big\|1_I\psi - 1_{I_n}\psi\big\| \le \big\|1_{I\setminus I_n}(\psi - f)\big\| + \big\|1_{I\setminus I_n} f\big\| < \epsilon + M\sqrt{\ell_n},$$
which is less than 2ϵ for large n. Finally, since $U^{-1} = U^\dagger$, each $U^{-1}1_I U$ is an orthogonal projection. Normalisation, the intersection identity and additivity carry over, since $U^{-1}1_I U\, U^{-1}1_J U = U^{-1}1_I 1_J U$ and conjugation by U keeps sums, 0 and $\mathbb{1}$. Continuity on increasing intervals holds because $\big\|U^{-1}1_I U\psi - U^{-1}1_{I_n}U\psi\big\| = \big\|1_I U\psi - 1_{I_n}U\psi\big\|$. ∎

Self-check 5.1

1. For the distribution of Example 5.2 and $f(\lambda) = \lambda^2$, take the set $\{-1\}$ and the cell $(1, 2]$, each with the largest height that the condition $\sum_j c_j 1_{S_j} \le f$ allows. What lower sum do they give?
2. For $\psi = (u_1 + u_2)/\sqrt{2}$ and the box energy, find $F_\psi(2E_1)$, $\int \lambda \,\mathrm{d}\mu_\psi$ and $\int \lambda^2 \,\mathrm{d}\mu_\psi$.

5.2 The spectral theorem

In *Quantum Mechanics I*, Postulate 4.1, an observable is a self-adjoint matrix $A = \sum_a a\, E_a$. For an operator with a domain, the sum becomes an integral, and the domain becomes a condition on the distributions. Let E be a spectral measure and A an operator. We write $A = \int \lambda \, \mathrm{d}E(\lambda)$ when

$$D(A) = \left\{ \psi : \int \lambda^2 \, \mathrm{d}\mu_\psi(\lambda) < \infty \right\} \quad \text{and} \quad \langle \psi, A\psi \rangle = \int \lambda \, \mathrm{d}\mu_\psi(\lambda) \quad \text{for every } \psi \in D(A). \tag{5.2}$$

Since $|\lambda| \le 1 + \lambda^2$ and $\mu_\psi(\mathbb{R}) = \|\psi\|^2$ is finite, the second integral converges when the first does. For a unit vector ψ, the first condition says that the distribution μ_ψ has a finite mean square, and the second that $\langle \psi, A\psi \rangle$ is its mean.

Theorem 5.4 (The spectral theorem · *cited*)**.** *For every spectral measure E there is exactly one operator A with $A = \int \lambda \, \mathrm{d}E(\lambda)$. It is self-adjoint, and $\|A\psi\|^2 = \int \lambda^2 \, \mathrm{d}\mu_\psi(\lambda)$ for every $\psi \in D(A)$. Conversely, for every self-adjoint operator A there is exactly one spectral measure E_A with $A = \int \lambda \, \mathrm{d}E_A(\lambda)$.*

So $E \mapsto \int \lambda \, \mathrm{d}E$ matches the spectral measures one to one with the self-adjoint operators. The first claim is Hall, Props 10.1–10.3: the means $\langle \psi, A\psi \rangle$ on a dense domain fix the operator A. The converse is Hall, Thm 10.4. His measures live on the Borel subsets of a set that Lecture 6 calls the spectrum, and he gives them the value 0 on the sets outside it (his Def. 10.5). Amrein, Prop. 4.30, gives the uniqueness among all spectral families $E_A((-\infty, \lambda])$. By the domain in (5.2), a unit vector ψ lies in $D(A)$ exactly when its distribution μ_ψ under E_A has a finite mean square, and by the formula for $\|A\psi\|^2$ this mean square is $\|A\psi\|^2$, as for the box's energy in § 1.7.

Proposition 5.5 (Explicit spectral measures)**.**

1. An eigenbasis. *Let (e_k) and λ_k be as in Proposition 5.3, with the spectral measure E made from them, and write $c_k = \langle e_k, \psi \rangle$. The operator B with*

$$D(B) = \left\{ \psi : \sum_k \lambda_k^2 \, |c_k|^2 < \infty \right\}, \qquad B\psi = \sum_k \lambda_k c_k e_k,$$

   is self-adjoint, and $B = \int \lambda \, \mathrm{d}E(\lambda)$. If a self-adjoint operator A has $Ae_k = \lambda_k e_k$ for every k, then $A = B$.
2. Position and momentum. *On $L^2(\mathbb{R})$, $Q = \int \lambda \, \mathrm{d}E_Q(\lambda)$ with $E_Q(I) = 1_I$, and $P = \int \lambda \, \mathrm{d}E_P(\lambda)$ with $E_P(I) = F^{-1} 1_I F$.*

Proof.

1. The steps of the proof of Theorem 1.8 on density, symmetry and the adjoint apply word for word, with e_k and λ_k in place of u_n and E_n, so B is self-adjoint. Next, $\mu_\psi(I) = \sum_{k:\, \lambda_k \in I} |c_k|^2$ has the points λ_k with the weights $|c_k|^2$. So $\int \lambda^2 \, \mathrm{d}\mu_\psi = \sum_k \lambda_k^2 \, |c_k|^2$, and by Parseval's identity $\langle \psi, B\psi \rangle = \sum_k \lambda_k \, |c_k|^2 = \int \lambda \, \mathrm{d}\mu_\psi$. Finally, let A be self-adjoint with $Ae_k = \lambda_k e_k$, and let $\psi \in D(A)$. Since λ_k is real, the symmetry of A gives $\langle e_k, A\psi \rangle = \langle Ae_k, \psi \rangle = \lambda_k c_k$. By Parseval's identity, $\sum_k \lambda_k^2 \, |c_k|^2 = \|A\psi\|^2 < \infty$, so $\psi \in D(B)$ and $A\psi = B\psi$. Hence $A \subseteq B$, and since adjoints reverse inclusions (Proposition 0.5), $B = B^\dagger \subseteq A^\dagger = A$. So $A = B$.
2. By Proposition 5.3, E_Q and E_P are spectral measures. For E_Q, the distribution $\mu_\psi(I) =$

$\int_I |\psi(x)|^2\,\mathrm{d}x$ has the density $|\psi|^2$. So $\int \lambda^2\,\mathrm{d}\mu_\psi = \int x^2\,|\psi(x)|^2\,\mathrm{d}x$, which is finite exactly when $\psi \in D(Q)$, and $\langle\psi, Q\psi\rangle = \int x\,|\psi(x)|^2\,\mathrm{d}x = \int \lambda\,\mathrm{d}\mu_\psi$. For E_P, since F is unitary, $\mu_\psi(I) = \|1_I F\psi\|^2 = \int_I \left|\tilde\psi(p)\right|^2\,\mathrm{d}p$ has the density $\left|\tilde\psi\right|^2$, and the same computation with p and $\tilde\psi$ gives P (Definition 4.4). ∎

By the uniqueness in Theorem 5.4, E is the spectral measure of B, and E_Q and E_P are those of Q and P. On the Borel sets S, the extension of E_Q is multiplication by the indicator of S (Amrein, Example 4.9). For the box Hamiltonian, Theorem 1.8 gives the eigenbasis u_n, so

$$E_H(I)\psi = \sum_{n:\,E_n\in I} c_n u_n, \qquad c_n = \langle u_n, \psi\rangle\,,$$

and in particular $E_H(\{E_n\}) = |u_n\rangle\langle u_n|$. The spectral measure is made from an eigenbasis in the same way for the circle's momentum, with the eigenbasis f_k of Proposition 3.2, and for each P_θ, with the eigenbasis $e_{\theta,k}$ of Theorem 3.10.

So each self-adjoint operator A has exactly one spectral measure E_A, by the cited spectral theorem, and a unit vector lies in $D(A)$ exactly when its distribution under E_A has a finite mean square. For the position, the momentum and every self-adjoint operator with an eigenbasis we have found E_A explicitly.

Self-check 5.2

1. For which numbers λ is $E_{P_\pi}(\{\lambda\}) \neq 0$? Which projection is $E_{P_0}(\{0\})$?

Lecture 5, second part

Readings on intervals and their consequences

5.3 Reading an interval

The first part matched each self-adjoint operator with exactly one spectral measure. In this part we postulate that a reading of an interval takes its probability from this measure. We then work out what a reading does: it conditions the distribution of the observable read, and it can change the distribution of another observable. Finally, we show that the momentum with walls does not decide its readings, and that no state on the line has bounded supports in both position and momentum.

Postulate 5.6 (Readings on intervals). The observables of a quantum system are self-adjoint operators. Let A be an observable and ψ a unit vector. An ideal reading of "A lies in I", for an interval I, on the state of ψ answers yes with the probability

$$\mathrm{Prob}(A \in I) = \|E_A(I)\psi\|^2\,. \tag{5.3}$$

An answer that occurs has positive probability. If it is yes, the state immediately afterwards is the ray of $E_A(I)\psi/\|E_A(I)\psi\|$. If it is no, the state is the ray of $(\mathbb{1} - E_A(I))\psi/\|(\mathbb{1} - E_A(I))\psi\|$.

This is Born's and Lüders' rule of *Quantum Mechanics I*, Postulate 4.1, for the projection $E_A(I)$. The postulate contains the three earlier rules. For an observable with an eigenbasis, $E_A(\{a\})$ is the projection E_a of Postulate 1.5. On the line, Proposition 5.5 gives the projections 1_I and $F^{-1}1_I F$ of Postulate 4.5. In the box, the same proof on $L^2(0,L)$ gives $E_Q(I) = 1_{I\cap[0,L]}$ for the position Q of § 4.4. For $I \subseteq [0,L]$ this is the projection 1_I of Postulate 2.6.

In the state immediately after a yes on I, the distribution of A is its earlier distribution conditioned on I. Let $\varphi = E_A(I)\psi / \|E_A(I)\psi\|$. By the intersection identity of Definition 5.1, $E_A(J)E_A(I) = E_A(I \cap J)$, so for every interval J

$$\mathrm{Prob}_\varphi(A \in J) = \frac{\|E_A(I \cap J)\psi\|^2}{\|E_A(I)\psi\|^2} = \frac{\mathrm{Prob}_\psi(A \in I \cap J)}{\mathrm{Prob}_\psi(A \in I)}.$$

This is Bayes' rule for the event that A lies in I. A reading of another observable need not follow such a rule, as the example below shows.

A single point is an interval. Multiplication by the indicator of $\{x\}$ maps every ψ to a function that vanishes off the point x, which is the zero class. So $E_Q(\{x\}) = 0$. A position reading at a point has the probability 0 in every state, so its yes never occurs, and no state follows it. For the box's energy, by contrast, $E_H(\{E_n\}) = |u_n\rangle\langle u_n| \neq 0$.

Physics texts write $E_Q(I) = \int_I |x\rangle\langle x| \, \mathrm{d}x$. There is no vector $|x\rangle$ in $L^2(\mathbb{R})$, since a function that vanishes off the point x is the zero class. So $|x\rangle\langle x| \, \mathrm{d}x$ is notation for $\mathrm{d}E_Q(x)$, used only under an integral, where it stands for $\langle \varphi, E_Q(I)\psi \rangle = \int_I \overline{\varphi(x)}\, \psi(x) \, \mathrm{d}x$.

Figure 5.2 draws the Gaussian of Example 5.7 below and the state that a yes on the window $[-s, s]$ leaves, with their position and momentum densities and the momentum density of the second state multiplied by p^2.

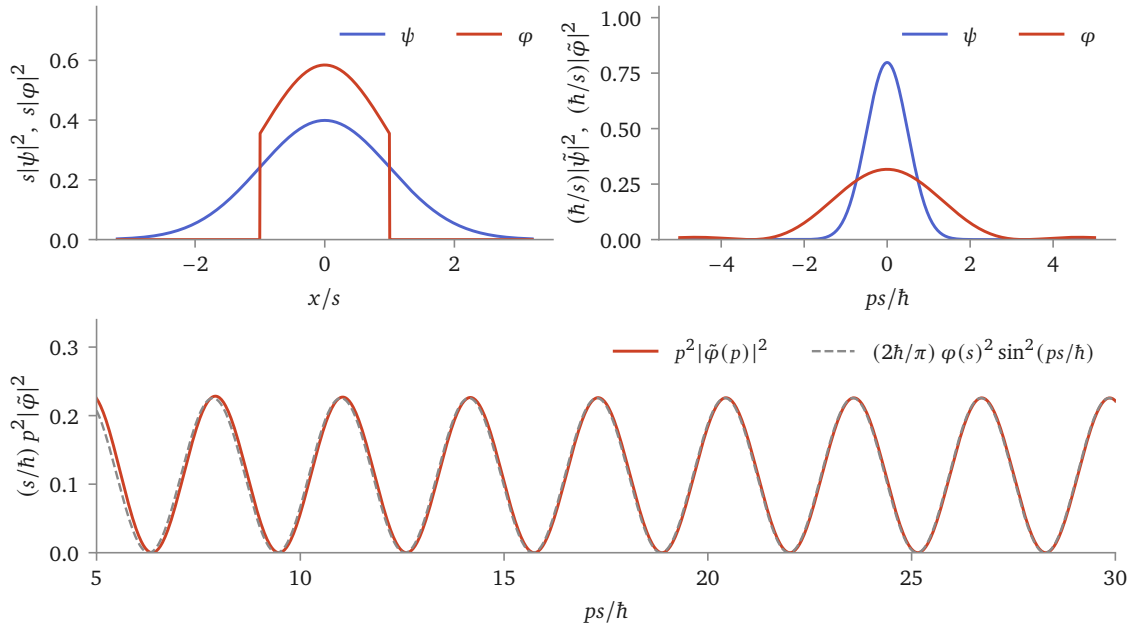


Figure 5.2: The Gaussian ψ (blue) and the state φ after a yes on $[-s, s]$ (red). Top left: the position densities. Top right: the momentum densities, on a scale where the tails of φ are too small to see. Bottom: $p^2\,|\tilde{\varphi}(p)|^2$ (red) with its boundary term $(2\hbar/\pi)\varphi(s)^2 \sin^2(ps/\hbar)$ (dashed). These keep their height as p grows.

Example 5.7 (A Gaussian read through a window)**.** Take the packet of Theorem 4.9 with $q = p_0 = 0$,

$$\psi(x) = (2\pi s^2)^{-1/4} e^{-x^2/4s^2},$$

and the window $I = [-s, s]$.

1. *The reading.* By (5.3) and Proposition 5.5, the position reads in I with the probability $c_I = \int_{-s}^{s} |\psi(x)|^2 \, \mathrm{d}x \approx 0.683$, the probability that a normal reading lies within one spread of its mean.
2. *The state after a yes.* By the postulate, it is the ray of $\varphi = 1_I\psi/\sqrt{c_I}$, the packet cut off at $\pm s$. At $\pm s$ the function φ jumps to 0 from $\varphi(s) = (2\pi s^2)^{-1/4} e^{-1/4}/\sqrt{c_I}$. By the conditioning rule, the position density of φ is $|\psi|^2 / c_I$ on I and 0 outside.
3. *The momentum before.* By Example 4.12, the momentum density of ψ is normal, with the mean 0 and the spread $\hbar/2s$.
4. *The momentum after.* Since φ vanishes outside I, its transform is the integral (4.1). This formula is part of Plancherel's theorem, the standing fact S1. Indeed, Hall, Thm A.19, computes $F\varphi$ as the limit of the integrals over $[-R, R]$, and for φ these do not change once $R \geq s$. So

$$\tilde{\varphi}(p) = \frac{1}{\sqrt{2\pi\hbar}} \int_{-s}^{s} e^{-\mathrm{i}px/\hbar} \varphi(x) \, \mathrm{d}x,$$

   an error function of a complex argument, computed numerically. A momentum reading lies in $[-a, a]$ with the probabilities

a	$\hbar/2s$	$\hbar/s$	$2\hbar/s$	$10\hbar/s$
before, ψ	0.683	0.954	1.000	1.000
after, φ	0.309	0.572	0.872	0.976

   The reading of the window has spread the momentum readings far beyond $\hbar/2s$.
5. *The boundary term.* For this cut Gaussian, the jumps at $\pm s$ leave a boundary term of order $1/p$ in the transform. Its square, multiplied by p^2, oscillates without decaying, as the bottom panel of Figure 5.2 shows. Two integrations by parts give the boundary term and a small remainder. Let $p \neq 0$. Since φ is even, the first integration by parts gives

$$\int_{-s}^{s} e^{-\mathrm{i}px/\hbar} \varphi(x) \, \mathrm{d}x = \frac{2\hbar}{p} \varphi(s) \sin\frac{ps}{\hbar} - \frac{\mathrm{i}\hbar}{p} \int_{-s}^{s} e^{-\mathrm{i}px/\hbar} \varphi'(x) \, \mathrm{d}x.$$

   The second bounds the last integral by $(\hbar/|p|)\big(2\,|\varphi'(s)| + \int_{-s}^{s} |\varphi''| \, \mathrm{d}x\big)$. On $[-s, s]$ we have $\varphi'' = (x^2/4s^4 - 1/2s^2)\varphi < 0$, so $\int_{-s}^{s} |\varphi''| \, \mathrm{d}x = 2\,|\varphi'(s)|$, and $\varphi'(s) = -\varphi(s)/2s$. So the bracket is $2\varphi(s)/s$. Dividing by $\sqrt{2\pi\hbar}$ and writing $\alpha = (2\hbar/\pi)^{1/2}\varphi(s)$, we get

$$\tilde{\varphi}(p) = \left(\frac{2\hbar}{\pi}\right)^{1/2} \varphi(s) \, \frac{\sin(ps/\hbar)}{p} + r(p), \qquad |r(p)| \leq \frac{C}{p^2},$$

   with the constant $C = \hbar\alpha/s$. So for large $|p|$,

$$p^2 \, |\tilde{\varphi}(p)|^2 \approx \frac{2\hbar}{\pi} \varphi(s)^2 \sin^2\frac{ps}{\hbar},$$

   which oscillates and does not decay.
6. *An infinite spread.* The oscillation has zeros, so the divergence needs an estimate. We prove the stronger statement that even $\int |p| \, |\tilde{\varphi}(p)|^2 \, \mathrm{d}p$ diverges, and the infinite mean square follows from it. We have $|\sin(ps/\hbar)| \geq \frac{1}{2}$ on the intervals $J_n = [(n + \frac{1}{6})\pi\hbar/s, (n + \frac{5}{6})\pi\hbar/s]$, for integers $n \geq 0$, each of the length $2\pi\hbar/3s$. For $n \geq 2$, every $p \in J_n$ has $p \geq 4\hbar/s$, so $|r(p)| \leq C/p^2 \leq \alpha/4p$. Hence $|\tilde{\varphi}(p)| \geq \alpha/2p - \alpha/4p = \alpha/4p$ on J_n. Since $p \leq (n+1)\pi\hbar/s$ there,

$$\int_{J_n} p \, |\tilde{\varphi}(p)|^2 \, \mathrm{d}p \geq \frac{\alpha^2}{16} \int_{J_n} \frac{\mathrm{d}p}{p} \geq \frac{\alpha^2}{24(n+1)}.$$

Summing over $n \geq 2$ gives a tail of the harmonic series, which diverges. So $\int |p|\,|\tilde{\varphi}(p)|^2\,\mathrm{d}p = \infty$, and the momentum readings of φ have no mean. Moreover, since $p^2 \geq (4\hbar/s)p$ for $p \geq 4\hbar/s$, we get $\int p^2\,|\tilde{\varphi}(p)|^2\,\mathrm{d}p = \infty$. Hence $\varphi \notin D(P)$, and $\Delta P = \infty$ in the convention of § 2.6.

This is the line's form of the reading of § 2.6. There a sharp reading of position left an infinite spread of the energy, and here it leaves an infinite spread of the momentum. The momentum readings are not conditioned on the window, as the position readings are. The reading has changed them.

Self-check 5.3

1. For ψ of Example 5.7, a student writes the probability that the position reads in $[0, s]$ as $\int_0^s x\,|\psi(x)|^2\,\mathrm{d}x$. What does this integral compute, and what is the probability?
2. For $\psi = (u_1 + u_2 + u_3)/\sqrt{3}$, read the box energy in $[3E_1, 10E_1]$. Find the probability of a yes and the state after a yes. What does a further energy reading then give?

5.4 A formula need not decide the readings

We now ask why an observable must be self-adjoint. Theorem 5.4 gives a first answer. A rule for the readings of an observable is a spectral measure, and by the theorem the spectral measures match the self-adjoint operators. The momentum with walls P_D of Definition 3.5 is symmetric and not self-adjoint (Proposition 3.7). So no spectral measure has P_D as its integral. Instead, many such integrals extend P_D in the sense of Definition 0.4, and we say that P_D *lies inside* each of them.

Proposition 5.8 (The momentum with walls lies inside every $\int \lambda\,\mathrm{d}E_\theta$)**.** *For $\theta \in [0, 2\pi)$, let $\lambda_{\theta,k} = \hbar(\theta + 2\pi k)/L$, and let E_θ be the spectral measure made from the orthonormal basis $e_{\theta,k}$ of Theorem 3.10 and the numbers $\lambda_{\theta,k}$, as in Proposition 5.3. Then $P_\theta = \int \lambda\,\mathrm{d}E_\theta(\lambda)$ and $P_D \subseteq P_\theta$ for every θ, and $E_\theta \neq E_{\theta'}$ when $\theta \neq \theta'$.*

Proof. By Theorem 3.10, P_θ is self-adjoint, and $P_\theta e_{\theta,k} = \lambda_{\theta,k} e_{\theta,k}$ for every k. If a self-adjoint operator has such an eigenbasis, Proposition 5.5 says that it is the diagonal operator of that basis, and that this operator is $\int \lambda\,\mathrm{d}E_\theta(\lambda)$. So $P_\theta = \int \lambda\,\mathrm{d}E_\theta(\lambda)$. Again by Theorem 3.10, $P_D \subseteq P_\theta$. Finally, $E_\theta(\{\lambda\}) \neq 0$ exactly when λ is one of the $\lambda_{\theta,k}$. For $\theta \neq \theta'$ in $[0, 2\pi)$ the sets $\hbar(\theta + 2\pi\mathbb{Z})/L$ and $\hbar(\theta' + 2\pi\mathbb{Z})/L$ are disjoint, since $\theta - \theta'$ is not a multiple of 2π. So $E_\theta(\{\lambda_{\theta,0}\}) \neq 0 = E_{\theta'}(\{\lambda_{\theta,0}\})$. ∎

By the uniqueness in Theorem 5.4, E_θ is E_{P_θ}. For the ground state $u_1 \in D(P_D)$, Example 3.12 found that E_0 and E_π give different distributions with the same mean and the same mean square. So the rule $-\mathrm{i}\hbar\psi'$ with the walls fixes the mean and the mean square of the readings on $D(P_D)$, but not the readings themselves. By Proposition 3.11 and Theorem 5.4, the E_θ are all the spectral measures whose integral extends P_D. To read the momentum of a particle in a box, one must choose θ. A symmetric operator need not decide the probabilities of its readings, and a self-adjoint operator does. Lecture 7 gives a second answer, from evolution.

Self-check 5.4

1. For which θ is $E_\theta(\{0\}) \neq 0$? What is then the probability that the reading is 0 for u_1?

5.5 No state on the line has bounded supports in both position and momentum

A state on the line cannot have both its position and its momentum readings in bounded intervals. A function has *bounded support* when it vanishes outside a bounded interval, and the claim is that ψ and $\tilde{\psi}$ cannot both have bounded support. In *Quantum Mechanics I*, Thm 9.3, a nonzero vector of $\mathbb{C}^N$ and its discrete Fourier transform cannot both have few nonzero entries. On the line the answer is sharper.

Theorem 5.9 (No state on the line has bounded supports in both position and momentum). *Let $\psi \in L^2(\mathbb{R})$ be nonzero and vanish outside a bounded interval. Then $\tilde{\psi}$ vanishes on no interval of positive length. In particular, $E_P(J)\psi \neq 0$ for every interval J of positive length, and no nonzero ψ has both $E_Q(I)\psi = \psi$ and $E_P(J)\psi = \psi$ for bounded intervals I and J.*

Proof. Choose X with $\psi = 0$ outside $[-X, X]$. As in Example 5.7, the transform is the integral

$$\tilde{\psi}(p) = \frac{1}{\sqrt{2\pi\hbar}} \int_{-X}^{X} e^{-\mathrm{i}px/\hbar} \psi(x)\, \mathrm{d}x \quad \text{for every } p.$$

Fix p_0. For $|x| \leq X$, the exponential series

$$e^{-\mathrm{i}px/\hbar} = e^{-\mathrm{i}p_0 x/\hbar} \sum_{n=0}^{\infty} \frac{1}{n!} \Big(\frac{-\mathrm{i}(p - p_0)x}{\hbar} \Big)^n$$

has terms of modulus at most $(|p - p_0|\, X/\hbar)^n / n!$. So, for every such x, its remainder after N terms is at most the remainder ϵ_N of the series of $e^{|p-p_0|X/\hbar}$. By Cauchy–Schwarz on $[-X, X]$, $\int_{-X}^{X} |\psi| \leq \sqrt{2X}\ \|\psi\|$. So the integral of the remainder against ψ is at most $\epsilon_N \sqrt{2X}\ \|\psi\|$, which tends to 0, and

$$\tilde{\psi}(p) = \sum_{n=0}^{\infty} a_n (p - p_0)^n, \qquad a_n = \frac{1}{\sqrt{2\pi\hbar}} \frac{1}{n!} \Big(\frac{-\mathrm{i}}{\hbar} \Big)^n \int_{-X}^{X} x^n e^{-\mathrm{i}p_0 x/\hbar} \psi(x)\, \mathrm{d}x,$$

for every real p. So $\tilde{\psi}$ is a power series about every point, and it converges on the whole line. Now suppose that $\tilde{\psi}$ vanishes on an interval J of positive length. As a power series, $\tilde{\psi}$ is continuous, and by the first fact about $\mathcal{A}$ in § 3.2, a continuous function in the zero class on J vanishes at every point of J. Take p_0 inside J. Since $\tilde{\psi}$ vanishes on an interval around p_0, every derivative of $\tilde{\psi}$ at p_0 is 0. The nth derivative of the series at p_0 is $n!\, a_n$, so every a_n is 0. Since this series converges on the whole line, $\tilde{\psi} = 0$ there, and $\psi = 0$ by the standing fact S1, a contradiction.

The last two claims follow. First, $\|E_P(J)\psi\|^2 = \int_J |\tilde{\psi}|^2$, which is positive. Next, let a nonzero ψ have $E_Q(I)\psi = \psi$ and $E_P(J)\psi = \psi$ for bounded intervals I and J. Then ψ vanishes outside I, so $\tilde{\psi}$ vanishes on no interval of positive length. But $E_P(J)\psi = \psi$ says that $\tilde{\psi}$ vanishes outside J, and so on a half-line, a contradiction. ∎

So the state φ that the window of Example 5.7 leaves gives every interval of momenta of positive length a positive probability. A state on the line can have its position readings in a bounded interval, or its momentum readings, but not both. The Gaussian packets have neither, since their densities are positive everywhere. The theorem concerns the intervals that hold the readings, while Theorem 4.9 concerns their spreads.

Self-check 5.5

1. Let ψ vanish outside $[-X, X]$. Can $\tilde{\psi}$ vanish at a single point? Try $\psi = 1_{[-X,X]}/\sqrt{2X}$.

The physics

In 2009 Bakr, Gillen, Peng, Fölling and Greiner imaged single atoms of rubidium-87 held in an optical lattice, a pattern of laser light whose potential minima form a square grid. The grid had the period 640 nm, and their microscope resolved about 600 nm, so each image showed which sites held an atom. In 2010 Sherson, Weitenberg, Endres, Cheneau, Bloch and Kuhr did the same in a lattice of period 532 nm. For a single atom, the question "is the atom on this site?" is a reading of its position on a square cell of the lattice, the two-dimensional form of a reading on an interval. Atoms on one site are lost in pairs during the imaging, so each image shows whether the number of atoms on a site was odd or even.

Readings on intervals

Observables: by the cited spectral theorem, every self-adjoint A is $\int \lambda \, \mathrm{d}E_A$ for exactly one spectral measure E_A, and $D(A)$ holds the vectors of finite mean square (Theorem 5.4). For Q, P and every self-adjoint operator with an eigenbasis, E_A is explicit (Proposition 5.5). The symmetric P_D lies inside the integral of every E_θ, so it does not decide its readings (Proposition 5.8). *Measurement:* a reading of "A lies in I" on a unit vector ψ answers yes with probability $\|E_A(I)\psi\|^2$, and a yes leaves the ray of $E_A(I)\psi/\|E_A(I)\psi\|$ (Postulate 5.6). A Gaussian read through a window leaves $\Delta P = \infty$ (Example 5.7). No nonzero state on the line has both its position and its momentum in bounded intervals (Theorem 5.9).

Summary.

- *Spectral measures*: projections on intervals with the defining properties, extended to the Borel sets by a cited theorem (Definition 5.1). An eigenbasis, the indicators on the line and their unitary images give spectral measures (Proposition 5.3).
- *The spectral theorem*: the spectral measures match the self-adjoint operators one to one, with $D(A)$ the vectors of finite mean square (Theorem 5.4, cited). The spectral measure is explicit for Q, P and every self-adjoint operator with an eigenbasis (Proposition 5.5).
- *Integrals against μ_ψ*: the distribution function F_ψ records μ_ψ, and the integral adds point weights and a density (Example 5.2).
- *Readings on intervals*: probability $\|E_A(I)\psi\|^2$, and a yes leaves $E_A(I)\psi/\|E_A(I)\psi\|$ (Postulate 5.6). A Gaussian read through a window leaves $\Delta P = \infty$ (Example 5.7).
- *The momentum with walls lies inside the integral of every E_θ*, so a symmetric operator need not decide its readings (Proposition 5.8).
- *No state on the line has bounded supports in both position and momentum*: a nonzero state that vanishes outside a bounded interval has a transform that vanishes on no interval of positive length (Theorem 5.9).

Lecture 6 locates the readings of an observable without a diagonalisation, in its spectrum.

Self-check answers.

5.1 1. The heights are $f(-1) = 1$ and 1, since $\lambda^2 > 1$ on $(1, 2]$ and comes as close to 1 as we like there. The lower sum is $1 \cdot \frac{1}{2} + 1 \cdot \frac{1}{2}(e^{-1} - e^{-2}) \approx 0.616$. 2. The weights are $\frac{1}{2}$ at E_1 and $\frac{1}{2}$ at $4E_1 = E_2$, so $F_\psi(2E_1) = \frac{1}{2}$, $\int \lambda \, \mathrm{d}\mu_\psi = \frac{5}{2}E_1$ and $\int \lambda^2 \, \mathrm{d}\mu_\psi = \frac{17}{2}E_1^2$.

5.2 1. Exactly for the eigenvalues $(2k+1)\pi\hbar/L$, $k \in \mathbb{Z}$, of P_π. The projection $E_{P_0}(\{0\})$ is $|f_0\rangle\langle f_0|$, on the constant functions.

5.3 1. It is the part of the mean position $\int x\,|\psi|^2 \, \mathrm{d}x$ that comes from $[0, s]$, a length, not a probability: $s(1 - e^{-1/2})/\sqrt{2\pi} \approx 0.157\,s$. The probability is $\int_0^s |\psi|^2 \, \mathrm{d}x = c_I/2 \approx 0.341$. 2. The interval contains E_2 and E_3, so $E_H([3E_1, 10E_1])\psi = (u_2 + u_3)/\sqrt{3}$. So a yes has the probability $\frac{2}{3}$, and the state after a yes is the ray of $(u_2 + u_3)/\sqrt{2}$, which is not an eigenvector. A further energy reading then gives $4E_1$ or $9E_1$, each with the probability $\frac{1}{2}$.

5.4 1. Only for $\theta = 0$, where $0 = \lambda_{0,0}$. Then $E_0(\{0\}) = |f_0\rangle\langle f_0|$, and the probability is $|\langle f_0, u_1\rangle|^2 = 8/\pi^2 \approx 0.811$.

5.5 1. Yes. For $\psi = 1_{[-X,X]}/\sqrt{2X}$, $\tilde{\psi}(p) = (\hbar/\pi X)^{1/2} \sin(pX/\hbar)/p$, which vanishes at $p = \pi\hbar/X$. A single point is not an interval of positive length.

References

Hall, *Quantum Theory for Mathematicians*, Props 10.1–10.3 (pp. 202–204), Thm 10.4 (p. 205), Def. 10.5 (p. 206), Thm A.19 (p. 532). Amrein, *Hilbert Space Methods in Quantum Mechanics*, §4.2.1 (pp. 141–143), Example 4.9 (p. 144), p. 168, Prop. 4.30 (p. 173). *Quantum Mechanics I*, Postulate 4.1 and Thm 9.3. Bakr, Gillen, Peng, Fölling and Greiner, *Nature* 462, 74 (2009). Sherson, Weitenberg, Endres, Cheneau, Bloch and Kuhr, *Nature* 467, 68 (2010).

Amrein, p. 168, states the probability of Postulate 5.6 in the same form.